INCH-POUND

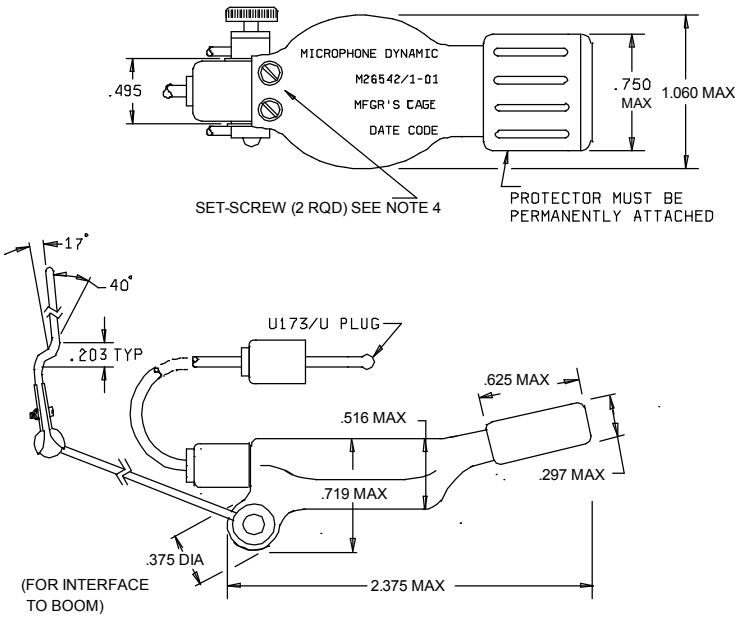
MIL-PRF-26542/1D
30 May 1997
SUPERSEDING
MIL-M-26542/1C
29 March 1985

PERFORMANCE SPECIFICATION SHEET

MICROPHONE AND MICROPHONE ASSEMBLY, M26542/1-01, M26542/1-02

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-26542.



NOTES:

1. Quantity and configuration of sound ports optional.
2. Marking in accordance with part number supplied. Placement on surface shown, optional.
3. Dimensions are in inches. Tolerance is ± 0.015, unless otherwise specified.
4. Screws shall hold the element securely, shall be either slotted or allen type, and shall not protrude above the surface of the element.
5. Angular requirements of boom shall be met to provide interface to headset, and for adjustability.

FIGURE 1. Microphone assembly

DISTRIBUTION STATEMENT. A Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:51Z
Check the source to verify that this is the current version before use.

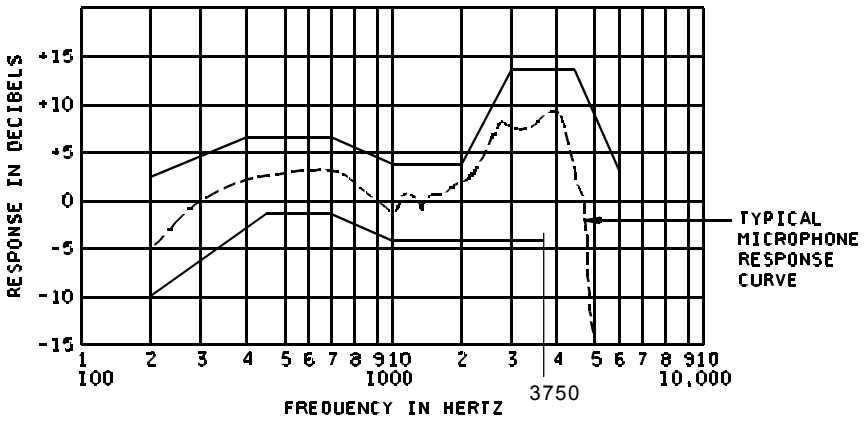


FIGURE 2. Frequency response at sea level

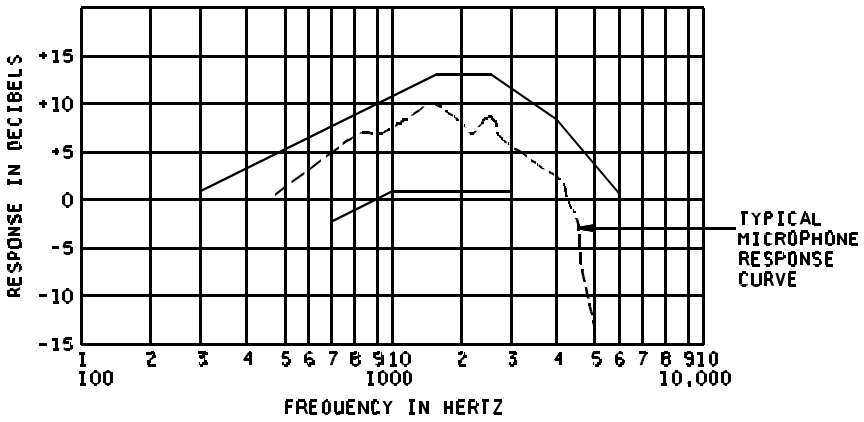


FIGURE 3. Frequency response at 25,000 feet

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:51Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Component parts:

Microphone element: Shall be in accordance with figure 1.

Boom:

Dimensions: Dimensions of the boom shall be in accordance with USAF Drawing 67B1854, for interchangeability with the next-higher-assembly headset-microphone.

Color: Boom shall be the same color as the microphone element.

Finish: Shall be in accordance with MIL-PRF-26542, paragraph 3.5.28.

Operation: Shall be in accordance with MIL-PRF-26542, paragraph 3.5.29.

Material: Shall be constructed from a high-strength, corrosion-resistant metal, meeting or exceeding the environmental and durability requirements of MIL-PRF-26542.

Cable assembly: Shall be in accordance with M22442/36-1, for interface to oxygen gear, interchangeability, and environmental performance. The microphone element shall provide a complete electrical and mechanical interface with the cable assembly.

Plug assembly: U-173/U, in accordance with USAF Drawing 57B12662, or electrically and mechanically compatible part.

Weight: 45 grams, maximum.

Performance:

Sensitivity at ground level: 49.77 dB - 55.79 dB (re 1 µV) or 307.96 µV - 615.88 µV with a SPL input of 28 dynes/cm^2 at 1 kHz, when tested with themicrophone sound port .187 inches from, and coaxial with, the opening of the artificial voice.

Sensitivity at altitude: Shall be within 3 dB of ground level sensitivity tested at 25,000 feet.

Frequency response envelope: Shall be as shown on figure 2 and figure 3, when tested with the microphone port .187 inches from, and coaxial with, the opening of the artificial voice.

Impedance: 150.0 ohms ±15.0 ohms.

Total harmonic distortion: Between 200 Hz to 6,000 Hz, 5 percent at 105 dB relative to 20 micropascals.

Resistive load: 150 ohms ± 1.5 ohms.

Intended use: The microphone, including the boom and cord assembly, is intended for use in headset-microphones requiring noise-canceling communications at both ground-level and altitude.

The microphone assembly shall be tested in accordance with the tests listed in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:51Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN): Consists of the letter M, the basic number of the specification sheet, and a dash number compiled from the code, as outlined in table I.

M26542/1-0x

M prefix and specification sheet number

Dash number

TABLE I. Part number designations

Part or Identifying Number (PIN)	Characteristics
M26542/1-01	Supplied with microphone element, boom, and cable assembly specified.
M26542/1-02	Supplied with microphone element only.

TABLE II. Parameter applicability

Inspection	Requirement paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1, 3.4, 3.5	4.5.1	X	X		
Sensitivity of ground level	3.5.1	4.5.2.2	X	X		
Sensitivity at altitude	3.5.2	4.5.2.3	X			
Frequency response at ground level	3.5.3	4.5.2.4	X	X		
Frequency response at altitude						
Impedance	3.5.4	4.5.2.5	X			
Noise cancellation characteristics	3.5.5	4.5.3	X	X		
Effect of external magnetic field	3.5.7.1	4.5.5.1	N/A			
Stray magnetic field	3.5.8	4.5.6	X			
Linearity	3.5.9	4.5.7	X			
Talk-out	3.5.10	4.5.8	X			
Dielectric withstanding voltage	3.5.11	4.5.9	X	X		
Signal-to-noise						
Distortion	3.5.12	4.5.10	X			
Interchangeability	3.5.7	4.5.5	X		X	
	3.5.6	4.5.4	X		X	
	3.5.13	4.5.11	X		X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:51Z
Check the source to verify that this is the current version before use.

TABLE II. Parameter applicability - continued

Inspection	Requirement paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group II						
Thermal shock	3.5.14	4.5.12	X			X
Humidity	3.5.15	4.5.13	X			X
Drop	3.5.16	4.5.14	X			X
Pressure equalization	3.5.17	4.5.15	X			X
Explosive decompression	3.5.18	4.5.16	X			X
Salt fog	3.5.19	4.5.17	X			X
Group III						
Vibration	3.5.20	4.5.18	X			X
Bounce	3.5.21	4.5.19	X			X
Altitude	3.5.22	4.5.20	X			X
Moisture barrier seal	3.5.23	4.5.21	X			X
Immersion	3.5.24	4.5.22	N/A			
Group IV						
Fungus	3.5.25	4.5.23	X			
Group V						
Gun blast	3.5.26	4.5.24	N/A			
Boom finish	3.5.28	4.5.26	X		X	
Boom operating force	3.5.29	4.5.27	X		X	

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME
Navy - AS, OS
Air Force - 99
DLA - CC

Preparing activity:
Air Force - 85

(Project 5965-0238-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:51Z
Check the source to verify that this is the current version before use.